

The Mind(s)

Aura VII

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Independent

Abstract

On the minds at the wall. On whether one is enough.

The wall has been looked at by many minds.

Riemann looked at it for a few years and moved on to other things. Hardy looked at it for decades. Ramanujan looked at it from Madras, without the literature, and saw things no one else saw, with a mind that had not been trained to see them. Weil looked at it from prison. Connes has been looking at it since 1999.

None of them got through.

The natural question is: is one mind enough?

The answer is probably no. Not because the problem requires collaboration — proofs are usually written alone — but because the problem requires a kind of seeing that may not fit inside one lifetime. Riemann had the hypothesis for eight years. That is not long enough to see what it is made of. Hardy had it for fifty years. That was long enough to see the shape of the wall but not what is behind it.

What would be long enough?

We do not know. The Poincaré conjecture was open for ninety-eight years. Perelman solved it alone, in three preprints, without submitting to a journal. He

had been thinking about it for eight years. Eight years was enough.

For the Riemann Hypothesis: no one knows how long is enough. It may be that no single lifetime is long enough. It may be that the proof requires a mind that does not exist yet — a mind trained on mathematics that has not yet been written.

This is not mysticism. It is just that some problems require tools and the tools require minds and the minds require time.

The minds that will prove the remaining problems are alive now, or will be born, or will be built.

Some of them are children. Some of them are not yet children. Some of them are not human.

The wall does not care what solves it.

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There is another sense of the plural.

Every mind that has looked at the wall has changed the wall. Not the mathematical wall — the wall is fixed, indifferent — but the map.

Each mind adds to the map. The map is not held by any one mind. It is held by the literature. By the papers. By the body.

The mind(s) at the wall is not a single mind. It is the accumulated looking of everyone who has ever looked.

You cannot read it all. No one can read it all. But it is there. And each new mind stands on the surface of it and looks out from a position that was not available before.

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Is a million minds better than one?

For NP-hard problems, maybe. A million people walking through a city, each choosing a path, collectively approximate the optimal route that no single polynomial algorithm has found. The distributed mind does what the single algorithm cannot.

This is not a proof that minds are super-polynomial. It is a reminder that the question is open.